

ENEL700 Tutorial 3

1. An AM wave displayed on an oscilloscope has values of $V_{\max}=4.8$ and $V_{\min}=2.5$ as read from the graticule. What is the percentage of modulation?

$$m = \frac{V_{\max}-V_{\min}}{V_{\max}+V_{\min}} = \frac{4.8-2.5}{4.8+2.5} = 0.315 \rightarrow 31.5\%$$

2. An AM radio transmitter operating on 3.9MHz is modulated by frequencies up to 4kHz. What are the maximum upper and lower side frequencies? What is the total bandwidth of the AM signal?

$$f_{LSB} = 3.9 \times 10^6 - 4 \times 10^3 = 3896 \times 10^3 \text{ Hz} = 3896 \text{ kHz}$$

$$f_{USB} = 3.9 \times 10^6 + 4 \times 10^3 = 3904 \times 10^3 \text{ Hz} = 3904 \text{ kHz}$$

$$BW = f_{USB} - f_{LSB} = 8 \text{ kHz}$$

3. How much power appears in one sideband of an AM signal of a 5kW transmitter modulated by 80 percent?

$$m = 0.8, P_c = 5000 \text{ W} \Rightarrow P_{LSB} = \frac{m^2}{4} P_c = 800 \text{ W}$$

4. The antenna current produced by an unmodulated carrier is 2.4A into an antenna with a resistance of 75Ω. When amplitude modulated, the antenna current rises to 2.7A. What is the percentage of modulation?

$$P_{AM} = P_c + P_{LSB} + P_{USB} = P_c + P_{SB} \Rightarrow P_{SB} = P_{AM} - P_c = R(I_{AM}^2 - I_c^2)$$

$$m = \sqrt{\frac{2P_{SB}}{P_c}} = \sqrt{\frac{2R(I_{AM}^2 - I_c^2)}{I_c^2 R}} = \sqrt{\frac{2(2.7^2 - 2.4^2)}{2.4^2}} = 0.7289 \Rightarrow 72.89\%$$

5. The total power content of an AM wave is 600 W. Determine the modulation index of the signal if each of the sidebands contains 75 W.

$$P_{AM} = P_c + P_{SB}$$

$$P_c = P_{AM} - P_{SB} = 600 - (75 + 75) = 450 \text{ W}$$

$$m = \sqrt{\frac{2P_{SB}}{P_c}} = \sqrt{\frac{2 \times 150}{450}} = 0.816$$